



Anti-apoptosis effect of traditional Chinese medicine in the treatment of cerebral ischemia–reperfusion injury

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Abstract

Cerebral ischemia, one of the leading causes of neurological dysfunction of brain cells, muscle dysfunction, and death, brings great harm and challenges to individual health, families, and society. Blood flow disruption causes decreased glucose and oxygen, insufficient to maintain normal brain tissue metabolism, resulting in intracellular calcium overload, oxidative stress, neurotoxicity of excitatory amino acids, and inflammation, ultimately leading to neuronal cell necrosis, apoptosis, or neurological abnormalities. This paper summarizes the specific mechanism of cell injury that apoptosis triggered by reperfusion after cerebral ischemia, the related proteins involved in apoptosis, and the experimental progress of herbal medicine treatment through searching, analyzing, and summarizing the PubMed and Web Of Science databases, which includes active ingredients of herbal medicine, prescriptions, Chinese patent medicines, and herbal extracts, providing a new target or new strategy for drug treatment, and providing a reference for future experimental directions and using them to develop suitable small molecule drugs for clinical application. With the research of anti-apoptosis as the core, it is important to find highly effective, low toxicity, safe and cheap compounds from natural plants and animals with abundant resources to prevent and treat Cerebral ischemia/reperfusion (I/R) injury (CIR) and solve human suffering. In addition, understanding and summarizing the apoptotic mechanism of cerebral ischemia–reperfusion injury, the microscopic mechanism of CIR treatment, and the cellular pathways involved will help to develop new drugs.

Keywords Traditional Chinese medicine · Cerebral ischemia–reperfusion injury · Apoptosis · Molecular mechanisms

Introduction

Ischemia is the cause of approximately 87% of strokes, and ischemic stroke is the third leading cause of death and the leading cause of disability in the worldwide, with a mortality rate of 30% [1, 2]. Pathologically, CIR causes cytoplasmic and chromatin condensation, nuclear fragmentation, the formation of membrane vesicles and membrane-bound apoptotic vesicles. The delayed cell death mode “programmed cell death (PCD)” that happens after CIR can provide an extended therapeutic window, which is an important

direction of research in ischemic stroke, where PCD is the most important and the first type of cell death to be identified [3].

Both ischemia and reperfusion due to interruption of cerebral blood flow cause brain cell damage and are part of a cascade of responses including energy barriers, acidosis, increased release of excitatory amino acids, formation of free radicals, and expression of apoptotic genes leading to apoptosis, necrosis, or other outcomes [4]. Besides apoptosis is reversible within hours after cerebral ischemia [5], so it is necessary to discuss pharmacological measures to reverse or intervene in the apoptotic pathway.

Mechanical embolization and intravenous infusion of fibrinogen activator (tPA) are frequently used clinically to restore ischemic tissue; however, secondary pathological damage to brain cells frequently occurs after blood reperfusion to the ischemic site, and tPA has its own complications, including intracranial hemorrhage, blood–brain barrier damage, and neurotoxicity [6]. TPA must be administered within 4.5 h after the beginning of a stroke for it to

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be therapeutically effective [7]. Reperfusion beyond the time frame may cause ischemia/reperfusion damage, enhanced hemorrhagic transformation, and impairment of cerebral function in ischemic stroke models. It has been shown, however, that reperfusion within 6 to 24 h may also restore ischemic tissue without significant ischemia/reperfusion harm and hemorrhagic transformation [8].

Small molecule medications, which include natural, synthetic, and endogenous substances used for the treatment and prevention of many illnesses, play a significant role and have become a key source for the discovery and development of novel therapies, which is favored by scientists. More than 66% of FDA-approved medications in 2020 will be small-molecule pharmaceuticals. After combining Traditional Chinese medicine with current scientific research, the fast rise of experimental research on small molecules in Chinese medicine plays a major role in the treatment of better clinical recovery, improved neurological function, and quality of life in stroke patients [9].

Inhibition of apoptosis by inhibiting cysteine aspartate protease may expose necrotic pathways, and similar backup systems may function when other kinds of cell death are suppressed [10]. Effective molecules of herbal medications may block one biological route to determine if herbal medicines interact with other pathways. The analysis stated that researchers are increasingly appreciating herbal medications and their natural active components due to their potential effects on healing the physiological central nervous system [11]. This publication also provides

an overview of the apoptosis-related proteins and their respective pathways, as well as an introduction to the three primary apoptosis routes (the endogenous pathway, the exogenous pathway, and the Endoplasmic Reticulum Stress (ERS) pathway) [12]. To explore and summarise the protective effect of traditional Chinese medicine (TCM) on CIR through apoptosis and related mechanisms, we searched PubMed, the Web of Science database, for animal experiments on CIR, using the search terms “apoptosis” and “cerebral ischemia–reperfusion injury”, and conducted a review that will help further the development of drug candidates and provide information based on literature citations for use in clinical trials for CIR.

Apoptosis classical pathway

There are two pathways of apoptosis, including exogenous and endogenous, where exogenous is mediated by death receptors while endogenous is produced by organelles mitochondria and causes caspase activation, and endoplasmic reticulum is also involved in apoptosis, the process of which will be described in detail next, the extrinsic and intrinsic apoptotic pathway plays a critical role in apoptosis following IR (Fig. 1).

Death receptors, such as TNFR, DR4/DR5 and FASR, can be stimulated by apoptotic signals to activate DISC which are composed of FADD and TRADD. AIF sheared by

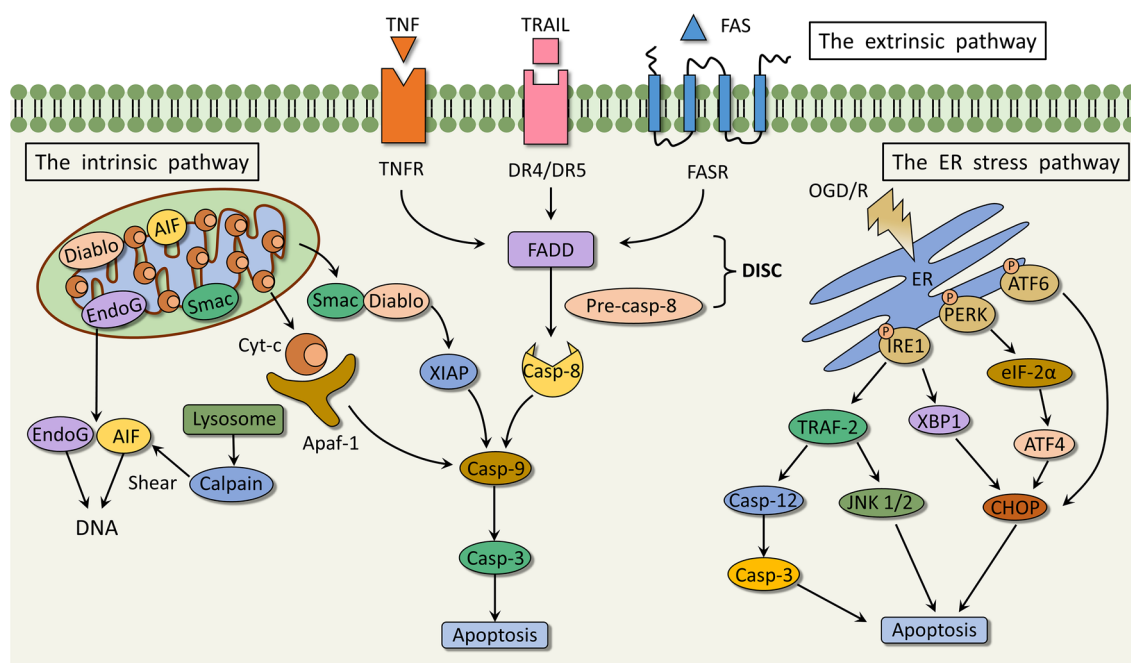


Fig. 1 Mechanism of extrinsic and intrinsic apoptosis pathway after CIR injury

calpain that released from lysosome and endogenous dopamine (EndoG) which are from mitochondria are sheared DNA. And then cause apoptosis cascade reaction.

Intrinsic pathway

Intrinsic apoptosis is defined as a kind of planned cell death that is initiated by intracellular or external stimulation and then produced by caspases (primarily caspase-3) as executor caspases, as proposed by the Nomenclature Committee on Cell Death (NCCD) [13].

The Bcl-2 family and the caspase family are mitochondrial upstream and downstream protein families, respectively. There are 14 members of the caspase family; seven of them cause apoptosis (caspases-2, 3, 6, 7, 8, 9, and 10). Caspases 2, 8, 9, and 10 have an extended pro-domain role; they serve as promoters for upstream signals that then trigger downstream effector caspases like caspases-3, 6, and 7. About 20 members of the Bcl-2 protein family can be subdivided into three groups based on structural homology: proteins that inhibit apoptosis, like Bcl-2 and Bcl-xl; proteins that promote apoptosis, like Bax and Bak [14]; and BH3-only proteins (Bcl-2 homology domain only proteins, so named because they contain only the BH3 structural domain), like Bad, Bid, Bim, Noxa, and PUMA. Bcl-2-binding component 3 (BBC3, also known as PUMA Bcl2-L11) and BH3-interacting structural domain death agonists (BID) are two examples of BH3-only proteins [15].

Bcl-2 protein family-related proteins play a crucial role in apoptosis induction via mitochondria-mediated cystathionine-dependent and non-dependent pathways [16], with the former being stimulated by the binding of cytochrome c (CytC) released into the cytoplasm to apoptotic protease activator-1 (Apaf-1) to induce apoptosome formation, which then triggers the truncation of the cell cycle [17]. The latter involves the transport of the apoptotic protein AIF from the cytoplasm to the nucleus.

This pathway involves the mitochondrial membrane permeability transition (MPT), a rapid increase in membrane permeability, the opening of the mitochondrial permeability transition pore (mPTP), and the release of apoptotic execution factors like CytC, second mitochondria-derived caspases activator (Smac)/Diablo, apoptosis-inducing factor (AIF), and endonuclease G from the mitochondrial matrix [18].

By decreasing the production of pro-apoptotic proteins like Bax, CytC, and activated Caspase-3 and raising anti-apoptotic proteins like Bcl-2, Ershi-wei Chenxiang tablets might alleviate neurological impairments, shrink brain infarcts, and boost neuronal function. Moreover, ECP improves neurological impairments, decreases the size of brain infarcts, and increases neuronal activity via increasing CaMKII and decreasing ATF4 and c-Jun [19]. Mediating protection of injured nerves with decreased CytC,

Bax, caspase-3, and caspase-9, and increased Bcl-2 [20], Tiliandin inhibits Ca^{2+} /calmodulin-dependent protein kinase II (CaMKII)-dependent mitochondrial-mediated apoptosis and MAPK/NF- κ B signaling pathways, providing protection against neurological damage. Each of them inhibits mitochondrial apoptosis by controlling CaMK II. The Xiao-Xu-Ming decoction suppresses the activation of caspase-9 and caspase-3 in the cytoplasm, raises c-IAP1, lowers p53 and Bax levels, and enhances Bcl-2 [21]. It also prevents the release of CytC and Smac/Diablo from mitochondria.

Curcumin's poor stability and difficulties reaching the brain have slowed its advancement as a clinical therapeutic agent [22]. In a rat model of acute cerebral ischemia, curcumin loaded onto macrophage-derived exosomes (Ex) prevented ROS-mediated mitochondrial apoptosis and lowered caspase-3, caspase-9, and Bax, therefore reducing brain damage [23]. By lowering MEK/ERK/CypD-dependent MPTP opening, the release of CytC from mitochondria was inhibited, preventing apoptosis mediated by the mitochondrial route and minimizing cell damage [24].

Cyperus esculentus L. orientin's antioxidant capabilities and consequent suppression of caspase-3 activation may serve to protect against I/R damage by preventing oxidative stress-mediated neuronal apoptosis and reversing the CoCl₂-induced reduction in cell survival [25]. By suppressing the NLRP3 inflammatory vesicle signaling system, Qingnao Dripping Pills (QNDP) were shown to have neuroprotective benefits. Specifically, QNDP decreased Bad and caspase-3 expression and increased Bcl-xl expression [26]. Apoptosis was suppressed by stigmasterol, which also reduced Bax and caspase-3 expression and increased Bcl-xl [27]. Caspase-3 and Bax were both downregulated by *Crocus sativus*, and apoptosis was dramatically decreased. Moreover, VEGF was shown to be elevated in I/R brain tissue [28]. It has been shown that danshensu reduces levels of caspase-3 and Bax while increasing Bcl-2 [29]. During anti-apoptosis, eleutheroside E increased expression of the 5-hydroxytryptamine receptor 2C (Htr2c) gene, and Htr2c suppressed expression of caspase-3, -6, and -7 [30]. Figure 2 shows the specific mechanism.

Extrinsic pathway

NCCD proposes defining exogenous apoptosis as a type of RCD stimulated by the extracellular environment sensed by plasma membrane receptors and transmitted intracellularly via Caspase-8 mediation [13]. These death receptors are members of the tumor necrosis factor receptor (TNFR) gene superfamily and are located on the plasma membrane. Activation of the protease caspase-8 and, later, caspase-3 by the death signaling complex formed by it and ligands induces cell death in damaged cells via protein hydrolysis [31]. The death receptor family includes TNFR-1 (also

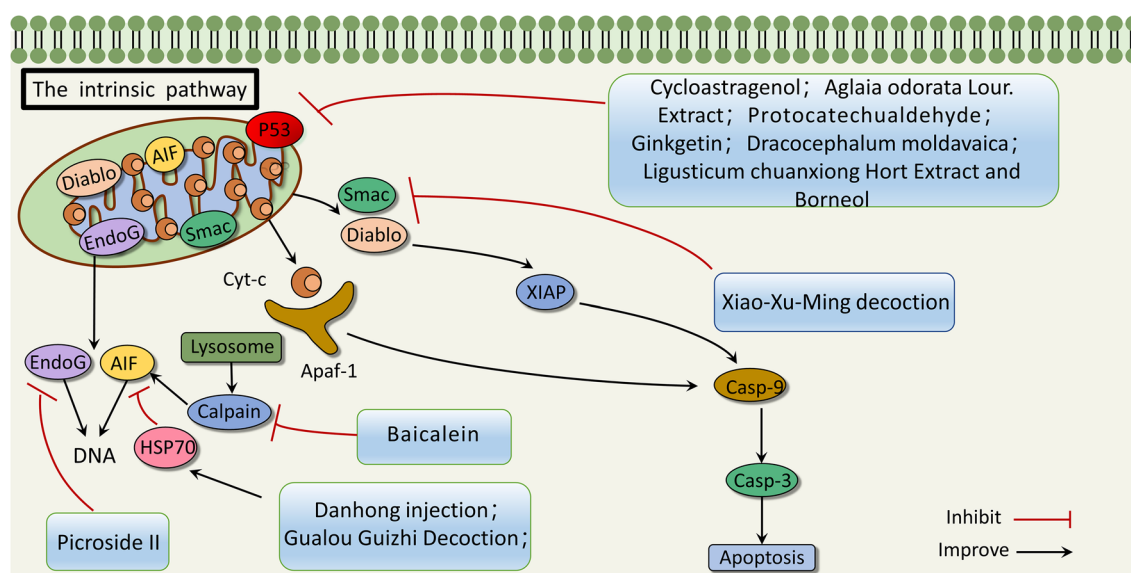


Fig. 2 Mechanism of Chinese medicine interfering with apoptosis of neuronal cells after CIR injury through the intrinsic pathway

known as DR1, CD120a, p55), CD95 (DR1, Fas, Apo-1), DR3 (also known as Apo-3, TRAMP, LARD), DR4 (TRAIL-R1, APO-2) and DR5 (TRAIL-R2) [32], and the death signalling complex formed by it and ligands activates the protease caspase-8 and later caspase-3, which

then induces cell in damaged cells through protein hydrolysis death. Figure 3 shows the specific mechanism.

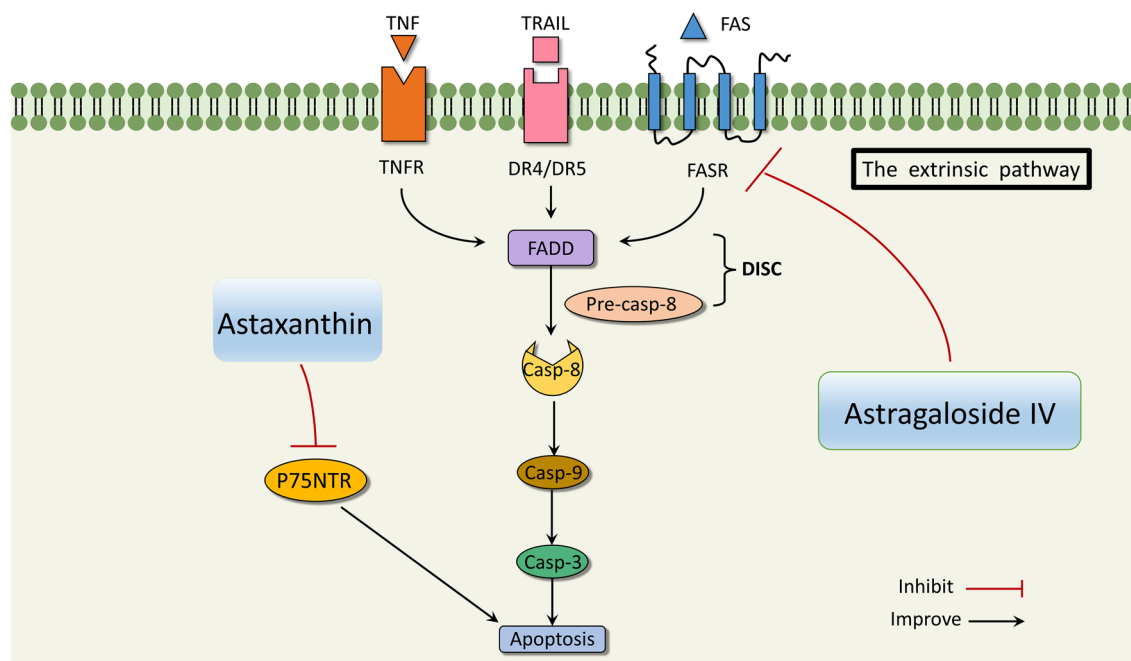


Fig. 3 Mechanism of Chinese medicine interfering with apoptosis of neuronal cells after CIR injury through the extrinsic pathway

P75 NTR

As a single-chain transmembrane protein with a molecular weight of 75 ku, neurotrophic factor receptor (p75NTR) is a member of the tumour necrosis factor receptor (TNFR) superfamily [33]. It is homologous to other receptors in the family and contains a death domain consisting of six helices, which is necessary to promote the pro-apoptotic function of the TNF receptor [34, 35]. Astaxanthin can lower p75NTR expression and the number of apoptotic cells [36].

Fas/FasL pathway

Fas is composed of a cytosolic C-terminal domain, a transmembrane region, and an extracellular N-terminal portion. A key component in the transmission of apoptotic signals is the death structural domain DD, which is located in the intracellular area [37]. Fas is a ligand of FasL, which is a type II transmembrane protein with three parts: an intracellular area, a transmembrane region, and an extracellular region. The intracellular portion is referred to as the death domain (DD). The ligand FasL, whose intracellular region can bind to a protein known as Fas-associated death domain protein (FADD), which contains a DD at the C-terminus that is homologous to Fas and therefore binds to each other, induces Fas to trimerize when the Fas-mediated death receptor pathway is activated. On the other hand, the death effect domain (DD) is the name of its N-terminal section. The death effector domain (DED) is the name given to the N-terminal end of FADD. After that, the N-terminal DED of FADD attaches to the caspase-8 zymogen to create the death signal-inducing complex DISC, which causes caspase-8 to become active. Caspases-8 and 3 may then be triggered downstream, which results in apoptosis [38–40]. Astragaloside IV slows down apoptosis by blocking the death receptor pathway and lowering the amount of mRNA for Fas, FasL, caspase-8, Bax, and Bcl-2 that are too high. Caspase-8, Bid, Caspase-3, and CytC, which are apoptotic cytokines, had less protein [41].

TRAIL/TRAILR pathway

Death receptors 4 (DR4) and 5 (DR5), two of the two kinds of TRAIL receptors, are able to transduce apoptotic signals and trigger apoptosis when attached to TRAIL. Recent years, however, have shown no herbal interventions in this route to cure CIR damage and hence reverse apoptosis.

Endoplasmic reticulum stress

Protein folding capacity of the endoplasmic reticulum (ER) can be disrupted by external or intracellular factors, leading to increased internal pressure and misfolded proteins. The ER is a central organelle located in the cytoplasm and

composed of membranes that synthesize, fold, and modify secreted and transmembrane proteins [42, 43].

Through the activation of activating transcription factor 6 (ATF6), inositol-requiring enzyme 1 α (IRE1 α), and PERK-like ER kinase (PERK), cells are able to detect endoplasmic reticulum stress and undertake a series of corrective reactions that help to preserve cellular homeostasis [44]. In normal settings, molecular chaperone binding immunoglobulin protein (BiP; also known as GRP78) binds to the aforementioned three proteins and maintains their inactive state. BiP also has an affinity for misfolded proteins; hence, it dissociates from the aforementioned proteins and activates the appropriate signal transduction pathways, such as: PERK-eukaryotic translation initiation factor 2 α (eIF-2 α)-ATF4-C/EBP homologous protein (CHOP), IRE1-tumour necrosis factor-associated factor receptor 2 (TRAF2)-activation and recruitment of apoptosis signal-regulated kinase 1 (ASK1) and ATF6 pathways [45]. When the unfolded protein response (UPR) is excessively severe or difficult to alleviate, it activates various pathways to trigger apoptosis, namely the PERK/eIF-2 α /activated transcription factor 4 (ATF4)/CHOP route, the c-JUN amino-terminal kinase (JNK) pathway, and the Caspase-12 pathway [46]. Figure 4 shows the specific mechanism.

IRE1 pathway

The cytoplasmic structural domain of IRE1 is phosphorylated, activating TRAF2, which then phosphorylates and activates ASK1, activating JNK, and triggering apoptosis [42]. By blocking the IRE1 α -XBP1 signaling pathway, icariin therapy lowers the production of XBP1u, XBP1s, cleaved caspase-3 mRNA and protein and reduces the p-IRE1 α /IRE1 α ratio, protecting neurons [47]. Longxuetongluo Capsule (LTC) lowers endoplasmic reticulum stress, inhibits activated MAPK, and suppresses the production of poly ADP-ribose polymerase (PARP), caspase-3, caspase-9, IRE1, CHOP, P38, and TRAF2, preventing PC12 apoptosis via the PERK/eIF-2 α and IRE1/TRAF2 pathways [48].

EIF-2 α pathway

To begin, PERK inhibits protein translation by phosphorylating eIF-2 α , and it increases the production of ATF4, which promotes cell survival by inducing the expression of ERS target genes and participating in redox processes [49, 50]. However, ATF4 may also boost CHOP expression and initiate apoptosis if it is active over an extended length of time.

(–)-Clausenamide prevented caspase-3 damage to neurons by reversing the activation of the GRP78/eIF-2 α -ATF4-CHOP branch, which inhibited ERS and apoptosis [51]. By reducing endoplasmic reticulum stress, downregulating

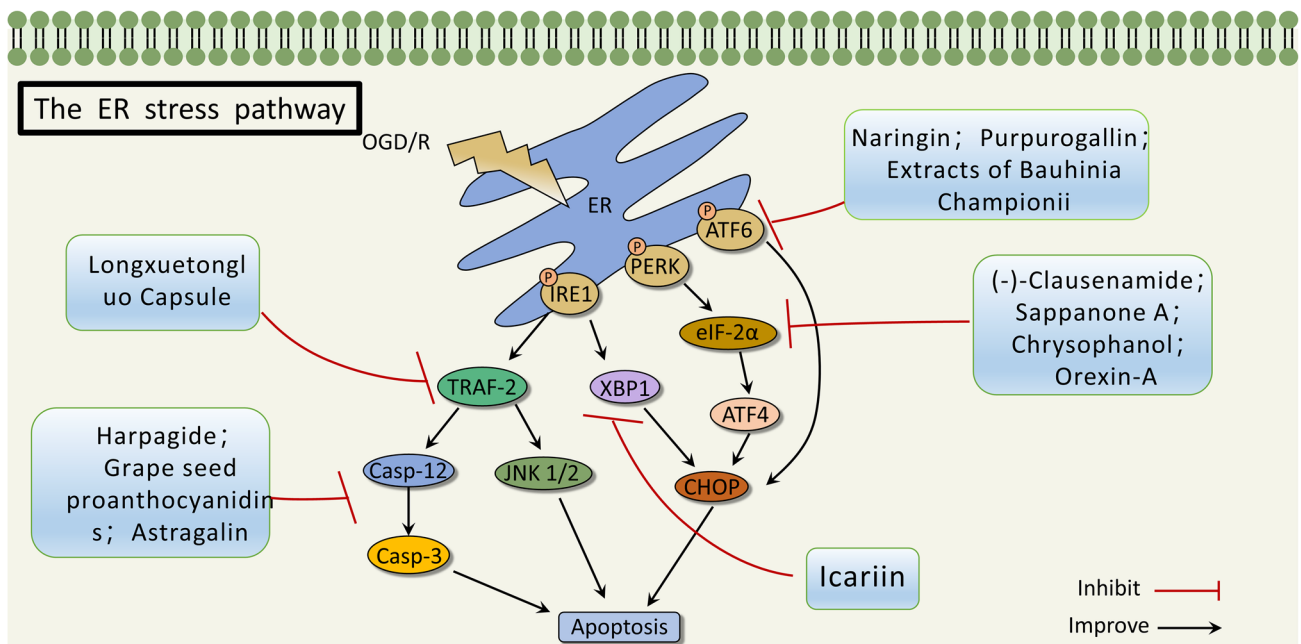


Fig. 4 Mechanism of Chinese medicine interfering with apoptosis of neuronal cells after CIR injury through the ERS pathway

CHOP, GRP78 (78 kDa glucose-regulated protein), eIF-2 α , Bax, caspase-3, CHOP and elevating Bcl-2, sappanone A protected brain tissue [52]. In the ischemic brain, chrysophanol increased I κ B- α and reduced GRP78, p-eIF-2 α , CHOP, and caspase-12 levels [53]. By preventing endoplasmic reticulum stress-mediated apoptosis, orexin-A lowered p-PERK, p-eIF-2 α , p-IRE1 α , p-JNK, caspase-12, CHOP, HSPA5, and the apoptotic protein caspase-3 to lessen damage [54]. Through this mechanism, harpagide also lowers GRP78, caspase-12, and CHOP [55]. By inhibiting GRP78 and caspase-12, grape seed proanthocyanidins (GSP) lessen endoplasmic reticulum stress-related apoptosis, and Bax and cleaved caspase-3 protein levels were considerably decreased. In contrast, p-ERK with caspase-12 was decreased and Bcl-2 levels were markedly elevated [56]. By limiting endoplasmic reticulum stress, minimizing excessive autophagy activation, and decreasing GRP78, CHOP, Bax, and caspase-3, berberine decreased apoptosis and enhanced cell survival in PC12 cells [57]. Isoginkgetin reduces p-JNK, Bax and Bim, TF4, CHOP, Atg5, Atg12, and P-JNK levels, while elevating Bcl-2 reduces apoptosis by regulating multiple ER-related signalling pathways [58]. Sodium formononetin-3'-sulfonate (Sul-F) inhibits ERS-mediated PERK/eIF-2 α /ATF4 and IRE1 signalling pathways in the ischemic penumbra, thereby reducing apoptosis [59]. Geraniol inhibits ERS via the PERK-ATF4-CHOP pathway. GRP78/PERK/ATF4 pathway in Gualou Guizhi Decoction reduces apoptosis by inhibiting ERS and upregulates eIF-2 α , SERCA

2 α [60]. Notoginsenoside R1 (NGR1) is a phytoestrogen that is isolated from *Panax notoginseng* [61]. NGR1 decreases CHOP, PERK, ERO1- α , IRE1- α , caspase-12, and elevates BCL-2 via ERS [62].

Intracellular if inward calcium flow leads to endoplasmic reticulum stress, causing lysosomes to release calpain, and calpain activation leads to TRPC6 degradation [63]. Naoluo Xintong capsule (NXC) reduces apoptosis and upregulates rCBF, lowers GRP78, CHOP and Caspase-12 expression, and attenuates neuronal death via activating TRPC6/MEK/CREB signaling pathway [64]. Astragaline (AST) elevated Bcl-2 through decreasing Bax and cleaved-caspase-3. In addition, AST reduced GRP78 and its downstream apoptotic mediators' CHOP and caspase-12 [65].

ATF6 pathway

By blocking the TLR4/NF- κ B pathway, purpurogallin upregulates MCL1, downregulates CHOP, ATF6 and GRP78, Bax, Bid, and Bim, and decreases endoplasmic reticulum stress and neuroinflammation [66]. Naringin, a compound that suppresses apoptosis through the ATF6 pathway, strongly reduces ERS-related proteins such GRP78, CHOP, Caspase-12, and ATF6 [67]. Extracts of *Bauhinia Championii* downregulate caspase-3, caspase-12, Bip, CHOP, ATF6, and Bax while upregulating Bcl-2 [68].

In the endoplasmic reticulum to the mitochondria transport process, Ca²⁺ is the basis for normal function

Table 1 Research advancements in using Chinese herbal medicine to treat CIR injury through the regulation of apoptosis

TCM	Model	Mechanisms	Targets	References
TQHXD	MCAO/R	ASK1/MKK4/JNK pathway	Reducing the protein levels of p-ASK1, caspase 3, p-MKK4, CytC, p-c-Jun, Bax/Bcl-2, and p-JNK; increasing the mRNA levels of Bcl-2	[128]
Tricin	OGD/R	PI3K/AKT pathway	Downregulating Beclin-1 protein expression, upregulating Bcl-2 protein, p-PI3K/PI3K and p-Akt/Akt expression	[100]
Platycodin D	OGD/R	PI3K/AKT/mTOR pathway	Increasing Bcl-2 expression; decreasing Bax expression	[102]
Notoginseng leaf triterpenes (PGL)	MCAO/R	MAPK and NF- κ B pathways	Reducing the concentrations of IL-6, TNF- α , IL-1 β , caspase 3/8/9 and HMGB1; down-regulating VCAM-1, MMP-9, MMP-2, and ICAM-1	[118]
Baicalein	MCAO/R	PI3K/AKT/mTOR pathway	Reducing phosphorylation of JNK, ERK, p38, caspase-3, I κ B α , Bcl-2 and nuclear translocation of NF- κ B/p65; decreasing Bax expression	[117]
Ginkgetin	MCAO/R	NF- κ B/p53 pathway	Decreasing Beclin-1 protein, p53 expression cathepsin B and cathepsin D; increasing Bcl-2 expression	[181]
Oroxylins A	OGD/R	Sonic Hedgehog pathway	Decreasing TNF α , MCP-1, and IL-1 β ; increasing Shh, p-smo and Gli2	[182]
Echinocystic acid	MCAO/R	JNK pathway	Upregulating the level of Bcl-2 and downregulating the levels of cleaved caspase-3 and Bax	[129]
Hederagenic acid	OGD/R	Nrf2/ARE pathway	Downregulating cleaved caspase-3 and Bax; upregulating pro-caspase-3, Bcl-2, Nrf2 and HO-1	[170]
<i>Dracocephalum moldevica</i>	MCAO/R	P53 pathway	Downregulating p53 and Bax; upregulating Bcl-2	[183]
Astaxanthin	OGD/R	Extrinsic pathway	Downregulating p75NTR protein expression	[36]
GLGZD	MCAO/R	PARP-1/AIF Pathway	Reducing PARP-1, AIF and Endo G	[184]
Senkyunolid-H	OGD/R	PI3K/AKT/NF- κ B pathway	Reducing NF- κ B, PI3K and cleaved caspase 3 enhanced expression of Bcl-XL	[103]
Resveratrol	OGD/R	AMPK-Mfn1 pathway	Downregulating Bcl-2 and surviving; upregulating Bax, Mfn 1 and caspase-9	[154]
Iridoid glycosides	MCAO/R	ERS pathway	Decreasing the expression levels of GrP78, CHOP and caspase-12	[73]
Lupeol	MCAO/R	PI3K/AKT pathway	Increasing the protein expression of p-PDK1, p-Akt, pc-Raf and p-BAD; downregulating the expression of cleaved caspase-3	[104]
Lupeol	MCAO/R	Nrf2 and P38 MAPK pathway	Decreasing caspase-3; inhibiting Bax/Bcl-2 and p38 MAPK phosphorylations	[116]
Crocin	MCAO/R OGD/R	Notch1 pathway	Increasing Notch1 and Bax; decreasing Bcl-2, NF- κ B, TNF- α and IL-6	[165]
Taraxasterol	OGD/R	Nrf2 pathway	Decreasing caspase-3 and Bcl-2; increasing Bax	[171]
Salvianolic acid D	MCAO/R OGD/R	TLR4/MyD88/NF- κ B pathway	Promoting Bcl-2 expression; suppressing Bax, CytC, cleaved caspase-3 and caspase -9; decreasing TLR4, MyD88, and TRAF6 proteins	[185]
Astragaloside IV	MCAO/R	Extrinsic pathway	Inhibiting the mRNA upregulation of Fas, FasL, Caspase-8, Bid, cleaved-caspase3, Cyto-C and Bax/Bcl-2	[41]
Theanine	MCAO/R	HO-1/ERK pathway	Decreasing IL-6, TNF- α , Bcl-2, HO-1 and IL-1 β ; increasing cleaved caspase-3, p-ERK1/2 and Bax	[142]

Table 1 (continued)

TCM	Model	Mechanisms	Targets	References
Aglaia odorata	MCAO/R	p53/Puma-mediated mitochondrial apoptosis pathway	Decreasing p53, cleaved caspase-9, cleaved caspase-3 and Puma	[186]
Lour. Extract	OGD/R			
Qingnao Dripping Pills	MCAO/R	NF- κ B pathway	Decreasing Bad, NLRP3, IL-1 β , IL-18 and cleaved caspase 3; increasing Bcl-xL	[26]
Astragaloside IV	OGD/R	JAK2/STAT3 pathway	Increasing p-JAK2 and p-STAT3; upregulating Bax and cleaved-caspase-3; downregulating Bcl-2	[150]
Danhong Injection	MCAO/R	PI3K/AKT pathway	Upregulating Bcl-2; downregulating Bad, Bax, p53 and Bim	[80]
Pachymic acid	MCAO/R	PI3K/AKT pathway	Upregulating the protein expression of p-PTEN, p-PDK1, p-Akt, pc-Raf and p-BAD; downregulated Cleaved caspase-3 protein expression	[86]
Curcumin	OGD/R	PI3K/AKT pathway	Decreasing caspase-3 and Bax; upregulating Bcl-2 and APE1	[87]
Sanggenon C	MCAO/R OGD/R	RhoA/ROCK pathway	Downregulating caspase-3, TNF- α , IL-1 β , IL-6 RhoA, ROCK1, ROCK2 and Bax; upregulating Bcl-2	[187]
Emodin	MCAO/R OGD/R	ERK-1/2 Pathway	Increasing Bcl-2 and GLT-1; decreasing activated-caspase 3	[139]
Polygala- saponin F	OGD/R	PI3K/AKT pathway	Upregulating the ratios of Bcl-2/Bax and p-Akt/total Akt; downregulating caspase-3	[85]
Stigmasterol	MCAO/R	Intrinsic pathway	Downregulating Bax and cleaved caspase-3; upregulating Bcl-XL	[27]
Kaempferol	MCAO/R	Nrf2 pathway	Decreasing TNF- α , IL-1 β and IL-6; down-regulating the expression levels of proteins NF- κ B, p-NF- κ B, and p-GSK-3 β ; upregulating Nrf-2 and p-Akt	[169]
Curcumin	OGD/R	TLR4/MyD88/MAPK/NF- κ B pathway	Decreasing TLR4, MyD88, p-NF- κ B P65, p-P38 MAPK, p-I κ B α , IL-1 β , IL-6 and TNF- α	[115]
Naringin	OGD/R	HIF-1 α /AKT/mTOR pathway	Decreasing NF κ B1 and Bax; Increasing Bcl-2	[188]
<i>Angelica sinensis</i> extract	MCAO/R	p38 MAPK/p90RSK/p-Bad pathway	Decreasing cleaved caspase-3, Cyt c and 4-HNE; Increasing p-CREB and BDNF	[114]
Eriocitrin	MCAO/R	Nrf2 pathway	Increasing p-Nrf2/Nrf2, NQO1 and HO-1; decreasing P65, IL-6, TNF- α , the mRNA levels of caspase-3 and caspase-9	[168]
Ligustilide	OGD/R MCAO/R	PI3K/AKT pathway	Decreasing the ratio of Bax/Bcl-2 and Cyt c	[89]
(-)-Clausenamide	OGD/R	eIF-2 α pathway	Decreasing GRP 78, p-eIF-2 α , ATF3, caspase-3 and CHOP	[51]
Asiaticoside	MCAO/R OGD/R	NOD2/MAPK/NF- κ B pathway	Decreasing Bax, Caspase-3; increasing Bcl-2	[189]
Anthocyanin	OGD/R	ASK1/JNK/p38 pathway	Decreased the protein levels of the phosphorylation of ASK1, JNK, Bax, Caspase-3 and p38; increasing Bcl-2	[120]
Irisin	MCAO/R OGD/R	Notch pathway	Decreasing Caspase-3, IL1 β and TNF- α ; increasing NICD, Notch1 and Hes 1	[164]
Ginkgetin	MCAO/R	PI3K/Akt/mTOR pathway	Increasing Bcl-2; Decreasing Bax, Caspase-3	[90]
Berberine	MCAO/R	CNPY2-PERK pathway ERS	Inhibiting CNPY2, GRP78, p-PERK/PERK, CHOP and Caspase-3	[75]
Resveratrol	MCAO/R	TGF- β -mediated ERK pathway	Decreasing Bad and Bax; Increasing Bcl-2 and Bcl-cl	[138]
Purpurogallin	MCAO/R OGD/R	TLR4/NF- κ B ERS pathway	Decreasing CHOP, ATF6, Bax, Bid, Bim and GRP78; Increasing MCL1	[66]

Table 1 (continued)

TCM	Model	Mechanisms	Targets	References
<i>Bauhinia Championii</i> (Extracts)	OGD/R	ERS pathway	Decreasing C-caspase-3, caspase-12, Bip, CHOP, ATF6 and Bax; Increasing Bcl-2	[68]
Icariin	OGD/R	IRE1/XBP1	Decreasing IRE1 α , p-IRE1 α , PERK, p-PERK, XBP1s, XBP1u, and caspase-3	[47]
Sappanone A	MCAO/R OGD/R	ERS pathway	Decreasing GRP78, eIF-2 α , Bax, caspase-3 and CHOP; Increasing Bcl-2	[52]
Harpagide	OGD/R	ERS pathway	Decreasing caspase-12, CHOP and GRP78	[55]
Astragalin	MCAO/R	ERS pathway	Decreasing Bax, GRP78, caspase-12, CHOP and cleaved-caspase-3; Increasing Bcl-2	[65]
Paeoniflorin and calycosin-7-glucoside	MCAO/R OGD/R	PI3K-AKT pathway	Increasing p-PI3K/PI3K, p-Akt/Akt and bcl-2/BAX; decreasing GSK-3 β	[94]
Chlorogenic Acid	MCAO/R OGD/R	PI3K-AKT pathway	Increasing p-PI3K/PI3K p-Akt/Akt and VEGFA; decreasing caspase-3	[79]
<i>Leonuri Herba</i> Total Alkali	MCAO/R	BDNF-TrkB-PI3K/AKT pathway	Inhibiting CD44, GFAP, FOXO1 and Bax; Increasing TrkB, p-PI3K, p-Akt, Bcl-2 and BDNF	[134]
Curcumin	MCAO/R	ERK-CHOP-caspase-11 pathway	Increasing caspase-11, ERK and CHOP; decreasing IL-1 β	[136]

between the endoplasmic reticulum and mitochondria and plays a vital role in cellular energy, autophagy, and apoptosis [69]. Sarcoplasmic Endoplasmic Reticulum Calcium ATPase (SERCA) is a calcium transporter protein present in the sarcoplasmic reticulum and endoplasmic reticulum, which serves to transport cytoplasmic Ca to the endoplasmic reticulum [70]. Salvinorin A inhibits elevated calcium levels and reactive oxygen species, and reduces apoptosis by downregulating GRP78, CHOP, and caspase-12 protein levels through the AMPK pathway to reduce endoplasmic reticulum stress [71]. Harpagide exerts neuroprotective effects through SERCA inhibition of endoplasmic reticulum stress, mainly regulating SERCA expression to maintain Ca²⁺ homeostasis, thereby reducing caspase-12 and CHOP, GRP78 [72].

Inhibiting apoptosis and ERS by downregulating GRP78, CHOP, and caspase-12, *Radix Scrophulariae* provides protection against CIRI in brain tissue [73]. Biochanin A reduces caspase-3, caspase-12, and Bax levels and raises Bcl-2 levels, all of which are protective mechanisms against cell death, to attenuate CIR damage [74]. By inhibiting the production of downstream proteins GRP78, p-PERK/PERK, CHOP, and Caspase-3, berberine protects neurons against the apoptosis caused by CIR damage through CNPY2 pathway [75].

Cellular pathways

PI3K/AKT pathway

The PI3K/AKT signaling pathway is composed of PI3K, a member of the lipid kinase family, and Akt, which is downstream. Based on its structure, PI3K, a phosphatidylinositol kinase that phosphorylates the third hydroxyl group of the inositol ring, can be divided into three classes, including class I heterodimers, which are heterodimers composed of a regulatory subunit (P85) and a catalytic subunit (P110), and class IA (PI3K α , β and δ) and class IB (PI3K γ) [76]. PKB, also known as Akt, a serine/threonine protein kinase with a relative molecular mass of approximately 6.0×10^4 , is central to this pathway, and three highly homologous isoforms exist as AKT1/PKB α , AKT2/PKB β and AKT3/PKB γ .

Ischemia drives cells to release chemicals that activate receptors with tyrosine kinase activity, such as nerve cell growth factor and platelet-derived growth factor (PDGF), and phosphorylate RTK through their tyrosine residues. The phosphorylated receptors then attach to the regulatory subunit of PI3K (P85), hence activating PI3K. When PI3K is turned on, it can phosphorylate inositol on the cell membrane to make a second messenger called phosphatidylinositol (3,4,5) trisphosphate (PIP3). PIP3 then binds to the PH domain of Akt, causing Akt to shift from the cytoplasm to the cytosol and changing the conformation of Akt. Akt then binds to phosphatidylinositol-dependent protein kinase 1 (PDK1) [77], which phosphorylates the Thr308 site and the Ser473 site of Akt. The activated Akt regulates apoptosis by directly phosphorylating apoptosis-related proteins

or indirectly altering the gene expression levels of apoptosis-related proteins, such as E2F, bcl-2 family/glycogen synthase 3 (GSK3), NF- κ B, Caspase-9 [78]. By modulating PI3K-Akt signaling, chlorogenic acid significantly prevented apoptosis and reduced MCAO-induced apoptosis, stimulated angiogenesis and elevated VEGFA levels, and lowered cleaved-caspase-3 [79]. The expression of p53 was downregulated, Bad, Bax, and Bim were upregulated, while Bcl-2 was upregulated after Danhong injection [80]. Inhibition of neuronal apoptosis and inflammatory responses were both brought about by Zhongfeng Capsules (ZFCs), with TLR 4, NF- κ B, Bax, and Caspase-3 being reduced and Bcl-2, p-PI3K, p-Akt, and I- κ B α being increased [81].

Sanhua decoction enhanced p-PI3K, p-Akt, and p-CREB1 levels and lowered TNF- α and IL-6 through modulating PI3K/Akt/CREB1 and TNF pathways [82]. Through modulating PI3K, AKT, and mTOR protein expression and inhibiting MDA, LDH, and Caspase-3, aloe-emodin provided protection through the PI3K/AKT/mTOR and NF- κ B signaling pathways [83]. By activating the PI3K/AKT/mTOR pathway, dehydrocostuslactone prevents apoptosis and autophagy, decreases PC12 cell damage, lowers Bax, Caspase-3, Caspase-7, and Caspase-9 expression, and dramatically raises Bcl-2, P-PI3K, P-Akt, and P-mTOR levels [84]. Polygalasaponin F (PGSF) may reduce apoptosis-induced damage via increasing PI3K/AKT signaling, since it increases Bcl-2/Bax expression and decreases caspase-3, Bcl-2/Bax, and the phosphorylated Akt/total Akt ratio [85]. Pachymic acid increases p-PTEN, p-PDK1, p-Akt, pc-Raf, and p-BAD while decreasing caspase-3 [86]. Curcumin has also been shown to enhance APE1 through the PI3K/AKT signaling pathway [87]. Through the PI3K/AKT/mTOR signaling pathway, daucosterol palmitate suppresses apoptosis and enhances PI3K, Akt, and mTOR [88]. Ligustilide reduces Bax/Bcl-2 ratio and suppresses Cyt c release and caspase-3 expression, whereas PI3K/AKT pathway activation increases Akt protein phosphorylation [89]. The prevention of apoptosis by ginkgetin attenuates the PI3K/AKT/mTOR signaling pathway by stimulating the downregulation of caspase-3 and Bax levels and the overexpression of Bcl-2 levels, hence increasing Akt and mTOR phosphorylation [90]. To maintain the integrity of the blood–brain barrier, exosome-like nanoparticles isolated and purified from bitter melon increased the ratio of Bcl-2 to Bax and suppressed caspase-3 by activating the PI3K/AKT pathway [91].

GSK-3 β is an important regulator of Bcl-2 protein family and promotes the occurrence of MOMP. Specifically, GSK-3 β phosphorylates Bax at Ser163, facilitating its translocation to mitochondria and the creation of apoptotic pores [92]. The SheXiang BaoXin Pill inhibits apoptosis by activating the PI3K/AKT pathway, as shown by the upregulation of p-Akt/Akt and p-GSK3 β /GSK3 β [93]. The combination of paeoniflorin and calycosin-7-glucoside elevates

p-PI3K/PI3K, p-AKT/AKT and Bcl-2/Bax and decreases GSK-3 β through the PI3K/AKT signalling pathway [94]. Avenanthramide C significantly upregulated p-PI3K, p-Akt, and p-GSK3 β , exerting neuroprotective and anti-apoptotic effects through activation of the PI3K/AKT/GSK3 β pathway [95]. Resveratrol inhibits neuronal apoptosis via the DJ-1/PI3K/AKT/GSK3 β pathway [96]. *Momordica charantia* exosome-like nanoparticles attenuate blood–brain barrier damage and inhibit neuronal apoptosis via upregulation of the AKT/GSK-3 β signalling pathway [97]. Oleuropein regulates the AKT/GSK-3 β signalling pathway to inhibit apoptosis [98].

By suppressing MAPK and stimulating the PI3K/AKT pathway, astilbin effectively suppresses apoptosis, reduces caspase-3 activation, lowers the Bax/Bcl-2 ratio and downregulates FADD [99]. Astilbin may decrease apoptosis and NLRP3 inflammasome inactivation. Tricin ameliorates learning memory impairment and lowers the extent of brain infarcts by downregulating inflammatory markers, LC3II/I and Beclin-1 through the PI3K/AKT pathway [100]. Through the PI3K/AKT/mTOR pathway, Platycodin D, a triterpenoid monomer isolated from the dried root of Platycodin [101], reduces Bax and increases bcl-2 [102]. Senkyunolid-H upregulates p-PI3K, p-Akt, and Bcl-xl protein levels while downregulating NF- κ B protein expression through activating the PI3K/AKT/NF- κ B signaling pathway [103]. Lupeol induced a decrease in caspase-3 expression while increasing that of p-PDK1, p-Akt, pc-Raf, and p-BAD via activation of PI3K/AKT [104]. Betulinic acid (BA), a naturally occurring pentacyclic lupane group triterpenoid and BA also regulate the PI3K pathway [105]. Salvianolic Acids inhibit apoptosis by upregulating mtCx43 through the PI3K/AKT pathway [106].

MAPK pathway

P38 MAPK pathway

Regulating apoptosis and inflammation after an ischemic stroke involves the MAPK and PI3K/AKT pathways [107]. It is known that mammals include four members of the MAPK family: P38 kinase (P38 α / β / γ / δ), extracellular signal-regulated kinase 1/2 (ERK1/2), c-Jun amino-terminal kinase (c-Jun N-terminal kinase, JNK1/2/3) and extracellular signal-regulated kinase 5 (ERK5) [108]. *Alpinia oxyphylla* Miq ameliorated apoptosis by downregulating caspase-3, while downregulating Smac/DIABLO, AIF/actin, p53/HSP60 and Bax/HSP60, but upregulating p-CREB/actin, XIAP/actin, Bcl-2/Bax and Bcl-xl/Bax, therefore the anti-mitochondrial-associated apoptotic effect can be attributed to up-regulated p38 MAPK/p90RSK-mediated p-Bad and CREB signaling versus down-regulated JNK/mediated Bax and p53 signaling [109]. Syringic acid regulates both JNK and p38 pathways

to reduce free calcium levels and reduce neuronal apoptosis [110]. Salidroside inhibits ASK1/MAPK and reduces CIR-induced apoptosis [111]. Diammonium Glycyrrhizinate increases Akt protein expression and decreases caspase-3, p38 MAPK and MMP-9 protein expression, thereby inhibiting apoptosis [112].

Through activation of p38 MAPK, phytic acid stimulates Bcl-2 expression, blocks Bax caspase3, and activates sestrin2 in PC-12 cells [113]. CytC/caspase-3-related signaling and p90RSK/CREB/BDNF signaling are both induced by *Angelica sinensis* extract, which acts via activating p38 MAPK-mediated p90RSK/p-Bad [114]. Down-regulating TLR4, MyD88, p-NF- κ B-P65, p-P38 MAPK, and p- κ B, as well as suppressing CCL3, are some of the mechanisms through which curcumin suppresses inflammation and promotes apoptosis [115]. The anti-apoptotic actions of lupeol, including the upregulation of the Nrf2 protein, the suppression of the Bax/Bcl-2 ratio, and the inhibition of caspase-3 activity and p38 MAPK activation, were initially described [116]. Baicalein reduces the production of IL-6, IL-18, and TNF- α because it blocks NF- κ B signaling by preventing the phosphorylation of I κ B α and the nuclear translocation of NF- κ B/p65. Additionally, the Bcl-2/Bax ratio was raised, and apoptosis was suppressed, by decreasing caspase-3 and upregulating JNK, ERK, and p38 activation [117]. Notoginseng leaf triterpenes (PNGL) enhanced nerve density, and PNGL (146 mg/kg) pretreatment suppressed NF- κ B expression and abolished the caspase 3/8/9 protein expression rise in p-P44/42, p-JNK1/2, and p-P38 MAPK levels. Thus, the protective effect may be attributable to the suppression of HMGB1-induced inflammation, inhibition of MAPK and activation of the NF- κ B pathway [118]. *Angelica Sinensis* promotes angiogenesis and suppresses neuronal apoptosis via the p38 MAPK/HIF-1/VEGF-A/vWF pathway [119].

Anthocyanin prevents apoptosis by blocking the ASK1/JNK/p38 signaling cascade via ASK1, p38, and JNK downregulation [120]. Sinensetin inhibits apoptosis by decreasing phosphorylation of p38, JNK, and ERK and activating the MAPK pathway [121]. MKK7 only activates JNK, whereas MKK4 stimulates both JNK and p38 MAPK [122]. Delavatine A lowers cell damage and death by inhibiting the MKK7/JNK signaling pathway, decreasing caspase-3, Bax, p-MKK 7 and p-JNK expression, and raising Bcl-2 [123]. The combination of Icaritin and Icariin reduced the expression of caspase-3, PARP, and Bax protein upregulation and decreases Bcl-2 from apoptosis, oxidative stress, lipid peroxidation, and extracellular matrix (ECM) accumulation [124]. Fraxin inhibits oxidative stress, inflammatory response, and apoptosis through PPAR γ activation, upregulates HO-1 and Nrf2 proteins, activates the Nrf2/HO-1 pathway, and inhibits in a dose-dependent manner NF- κ B, IKK- β p38 MAPK, ERK1/2

[125]. Cryptotanshinone inhibited neuronal apoptosis and down-regulated Caspase-3 activation and degradation of poly ADP ribose polymerase (PARP), as well as increased Bax/Bcl-2 ratio in neurons, and inhibited ERK1/2, JNK and p38 MAPK, suggesting an association with the MAPK signalling pathway [126].

JNK MAPK pathway

Hederagenin reduces p-JNK by inhibiting the MKK4/7-JNK axis, MLK3 phosphorylation, and activating NF- κ B and MEK4/7-JNK pathways [127]. Tong-Qiao-Huo-Xue decoction (TQHDX) generally decreases p-ASK1 by activating the ASK1/MKK4/JNK pathway to reduce p-ASK1, caspase 3, p-MKK4, CytC, p-c-Jun, Bax/Bcl-2, and p-JNK to protect against apoptosis [128]. Echinocystic acid (EA) provides neuroprotection through its anti-apoptotic and anti-inflammatory activity by inhibiting the JNK signalling pathway in brain I/R injury, upregulating Bcl-2 levels while downregulating caspase-3 and Bax and decreasing p-JNK [129]. Celastrol activates PI3K/AKT by downregulating AK005401 and MAP3K 12 protein, which increases PI3K and p-Akt/Akt ratios, improves antioxidant capacity, inhibits cleaved caspase-3, Bax/Bcl-2, p-JNK/JNK, and p-Jun/Jun ratios significantly [130]. Tetramethylpyrazine from the Chinese herb *Ligusticum sinense* ‘Chuanxiong’ inhibits the JNK signalling pathway, decreases C-fos, C-jun and P-JNK, and reduces apoptosis in hippocampal neurons [131].

ERK MAPK pathway

Upon binding of BDNF to TrkB, the intracellular domain of tyrosine residues is autophosphorylated, leading to ligand-induced dimerization of each receptor and subsequent activation of downstream cascade signals of three signalling pathways activated by TrkB: PI3K/AKT, MAPK/ERK, and PLC γ pathways, the first two of which are associated with apoptosis [132]. Baicalin increases TrkB, PI3K/AKT and MAPK/ERK, attenuated IR damage by regulating the BDNF-TrkB/PI3K/AKT and MAPK/Erk1/2 signalling axis in astrocytes, elevated p-CREB, BDNF-TrkB, p-ERK, p-Akt, and decreased Bax, caspase-3, and caspase-9 [133].

Leonuri Herba Total Alkali inhibits CD44, GFAP, and FOXO1 expression. It increases BDNF and NeuN. in addition, and it reduces apoptosis by upregulating TrkB, p-PI3K, p-Akt, Bcl-2, Bcl-2/Bax, and downregulating Bax through the BDNF-TrkB-PI3K/AKT signalling pathway [134]. Artemisinin inhibits apoptosis by activating the ERK1/2/CREB/Bcl-2 signaling pathway to inhibit apoptosis, resulting in reduced caspase-3, significantly increased P-ERK1/2 and P-CREB levels, significantly decreased Bax/Bcl-2 ratio, and in inflammation decreased inflammatory cytokines IL-18, IL-1 β and NF- κ B levels [135]. Curcumin inhibits apoptosis

through ERK-CHOP-caspase-11 pathway caspase-11, ERK and CHOP were increased while the expression of the inflammatory factor IL-1 β was decreased [136]. Tongxinluo was found to inhibit the decrease in p-MEK, p-ERK1/2, p-p90RSK and p-bad levels, the neuroprotective capacity of TXL, associated with a reduction in neuronal apoptosis and injury, so it can be assumed that both the PI3K/AKT pathway and the MEK1/2/ERK1/2/p90RSK pathway are involved in apoptosis [137]. Resveratrol regulates neuronal apoptosis by downregulating the transforming growth factor- β (TGF- β)-mediated ERK signalling pathway, with upregulation of Bcl-2 and Bcl-xl, downregulation of Bad and Bax, and an increase in the anti-inflammatory factors Iba-1 and MPO [138].

Emodin reduces apoptosis through ERK-1/2 and increases Bcl-2 and glutamate transporter-1 (GLT-1) expression but inhibits caspase-3 [139]. Echinochrome A (Ech A), a quinone pigment present in sea urchin shell spines [140], increases Bcl-2, p-ERK/ERK and p-Akt/Akt, BDNF expression levels, and decreased caspase-3 and Bax [141]. Theanine reduces oxidative stress and apoptosis by inhibiting HO-1 expression and activating the ERK1/2 pathway, and inhibits HO-1 expression and activity and increases ERK1/2 [142]. Calycosin is a natural product extracted from some plant families. Calycosin regulates the ERK1/2 pathway to inhibit apoptosis [143].

It has been demonstrated that activation of MEK/ERK triggers anti-apoptotic effects during cerebral ischemia [144]. Suppose phosphorylation of CREB is inhibited via the MEK/ERK pathway. In that case, it induces downregulation of B-cell lymphoma-2 (Bcl-2) family transcription and expression [145], curcumin alleviates neuronal apoptosis, increases p-MEK, p-ERK, p-CREB, and Bcl-2 and decreases Bax levels [146]. Rutaecarpine inhibited caspase-3 activation and Bax expression to suppress neuronal apoptosis, inhibited ERK1/2 phosphorylation and suppressed apoptosis by regulating ERK1/2 and Nrf2/HO-1 pathways [147].

JAK/STAT pathway

STAT3 is an inhibitor of neuronal apoptosis by inducing cell growth and repairing related genetic proteins such as apoptosis suppressor genes survivin, Bcl-2, Bcl-xl, and apoptosis regulator (cIAP2) [148]. Resveratrol upregulates PI3K/AKT/ by activating the JAK2/STAT3 pathway in the ischemic penumbra mTOR pathway, upregulates p-JAK2, p-STAT3, p-Akt, p-mTOR, Bcl-2, and decreases Bax and caspase-3 [149]. Astragaloside IV inhibits SH-SY5Y cell apoptosis, decreases Bax and caspase-3, and increases p-JAK2 and p-STAT3, Bcl-2 expression. AS-IV may protect the brain through the JAK2/STAT3 signalling pathway [150]. Nicotiflorin is a flavonoid derived from the *Carthamus tinctorius*.

It has been shown to reduce neuronal apoptosis through the JAK2/STAT3 pathway by downregulating p-JAK2 and p-STAT3 [5]. Puerarin inhibits apoptosis in astrocytes via the JAK2/STAT3 pathway [151]. Shaoyao Gancao decoction reduces apoptosis and promotes microglial phenotype shift from M1 to M2 via IL-13 and its downstream JAK2/STAT6 pathway [152].

AMPK pathway

Kaempferol mediates the AMPK signalling pathway to regulate autophagy and inhibit apoptosis, thereby protecting the brain from IR damage [153]. Mitofusin1 (Mfn1) maintains the mitochondrial membrane potential, ensuring that apoptosis-inducing factors within the mitochondria do not enter the cytoplasm to cause an apoptotic response. Resveratrol reduces oxidative stress, improves mitochondrial energy metabolism, and inhibits mitochondrial apoptosis, regulating its function through the AMPK-Mfn1 pathway [154]. Schaftoside improves cerebral ischemia by enhancing autophagy and reducing apoptosis and inflammation through the AMPK/mTOR pathway [155]. *Arctium lappa* L. roots are popular cultivated vegetables that exert anti-ischemic effects by inhibiting apoptosis and AMPK/mTOR-mediated autophagy [156]. Ligustilide is the main constituent of *angelica sinensis* and *Ligusticum sinense* ‘Chuanxiong’, which inhibits apoptosis by activating the AMPK signalling pathway to induce Drp1-mediated mitochondrial division [157].

Wnt pathway

The Wnt pathway is a downstream effector of the PI3K/AKT pathway, a classical anti-apoptotic signalling pathway in brain IR [158]. Curcumin reduces apoptosis and inflammation by inhibiting the Wnt/JNK1/NF- κ B signalling pathway, decreasing the expression levels of Wnt5a, IL-6, TNF- α , caspase-3 and phosphorylation of JNK1, and inhibiting NF- κ B [159]. 10-O (N, N-dimethylaminoethyl)-ginkgolide B methanesulfonate (XQ-1H), inhibits death and promotes neurogenesis in a Wnt/ β -catenin signalling-dependent manner, decreases GSK3 β /Caspase-3 activity, and enhances Wnt1/ β -catenin/Neuro D1/Cyclin D1 and Bcl-xl expression [160]. Dan-Deng-Tong-Nao capsules upregulated the expression of Bcl-2, β -catenin and Wnt-1, and downregulated the expression of Bax and GSK3 β , significantly increased the density of brain microvessels and exerted anti-neuronal apoptosis [161].

Notch pathway

Notch receptors contact and bind to Notch ligands on the surface of adjacent cells, eventually releasing the

intracellular domain (Notch intracellular domain, NICD), which subsequently enters the nucleus and binds to CSL-DNA binding proteins to recruit the nuclear transcription activator protein (mastermind-like, MAML) family, forming the NICD-CSL-MAML transcriptional complex, which activates downstream target genes Hes, Hey et al. [162]. NICD and HIF-1 α are synergistically involved in the pro-inflammatory and apoptotic signalling pathways in stroke [163]. Irisin reduces apoptosis in hippocampal neurons. In addition, it downregulated IL-1 β and TNF- α expression and upregulated NICD, Notch1 and Hes1 expression both in vivo and in vitro, and cleaved-caspase-3 was decreased after pretreatment [164]. Crocin promotes Jagged1, Bcl-2, Notch1, and decreases Bax by regulating the Notch signalling pathway, and increases rat proliferation and migration of neural stem cells and inhibits apoptosis of these cells [165].

Nrf2 pathway

By regulating the Nrf2/HO-1 pathway, geraniin protects against cerebral ischemia/reperfusion damage by decreasing oxidative stress and neuronal death. It also dramatically upregulated Nrf2 (total and nuclear proteins) and HO-1, suppressed Bax and caspase-3 expression, and upregulated Bcl-2 expression in a dose-dependent manner, resulting in an enhanced Bcl-2/Bax ratio both in vivo and in vitro [166].

Through the Nrf2-mediated NOX4/ROS/NF- κ B pathway, isoquercetin reduced oxidative stress and neuronal apoptosis following ischemia/reperfusion injury. It also downregulated caspase-3 protein expression and upregulated Bcl-2 protein expression. However, Nrf2 knockdown attenuated the anti-apoptosis of Iso in oxygen glucose deprivation reoxygenation (OGD/R) and inhibited the NOX4/ROS/NF- κ B signalling pathway [167]. The inflammatory response and apoptosis were suppressed by Nrf2/HO-1/NQO1/NF- κ B, as shown by the dose-dependent downregulation of caspase-3 and caspase-9 protein expression in the high and medium dosage groups of Eriocitrin, with no significant alterations found in the low dose group [168]. Kaempferol dramatically reduced I/R-induced neuronal apoptosis by decreasing expression of TNF- α , IL-1, and IL-6 in brain tissue, as well as downregulating expression of NF- κ B, p-NF- κ B, and p-GSK-3, and upregulating expression of Nrf-2 and p-Akt [169]. Hedera-genic acid lowers apoptosis by reducing caspase-3 activity, upregulating Bcl-2, and downregulating Bax to limit apoptosis, oxidative stress, and by activating the Nrf2/ARE pathway, inhibiting excessive ROS generation, lowering MDA levels, and boosting SOD activity [170]. By modulating the Nrf2 signaling pathway, taraxasterol protects against oxidative stress and apoptosis by decreasing caspase-3 activity and Bcl-2 and increasing Bax [171]. Genistein stimulates the Nrf2/HO-1 pathway and inhibits the NF- κ B pathway, reducing the levels of Bax and cleaved caspase-3. Additionally,

Bcl-2 levels were considerably elevated compared to the HI group [172]. Pratensein inhibits the activation of the NLRP3 inflammasome by activating Nrf2, thereby reducing the inflammatory response and apoptosis of HT22 cells [173]. Total Flavonoids in *Premna fulva Craib* (TFPFC) is a type of flavonoid compound extracted from *Premna fulva Craib* that reduces MDA levels, increases SOD activity, upregulates Nrf2 and HO-1, and inhibits apoptosis [174]. Diosmetin attenuates PC12 cell activity inhibition, apoptosis, activation of the Nrf2/ARE pathway mediated by Keap1 and inhibition of NLRP3 inflammatory vesicles [175]. Geraniin inhibits oxidative stress and neuronal apoptosis through the Nrf2/HO-1 pathway [176]. Pachymic acid contributes to increased cell viability and reduced apoptosis by targeting the Nrf2/HO-1 pathway and inhibiting miR-155 [177]. *Chrysanthemum morifolium* significantly reduces apoptosis and oxidative stress by regulating the Keap1/Nrf2 pathway [178]. Albiflorin inhibits apoptosis and inflammation by regulating the Nrf2/HO-1 pathway [179]. Tanshinone IIA reduced infarct volume, attenuated apoptosis and upregulated Nrf2 mRNA expression and protein levels via antioxidant and neuroprotective effects [180].

Other channels and pathways

The anti-inflammatory mechanism of Salvianolic Acid D may be attributed to the decrease of HMGB1 nucleoplasmic translocation and the suppression of its downstream TLR4/MyD88/NF- κ B signaling, increasing the expression of Bcl-2 and decreasing the expression of Bax, cytc and caspase-3 and -9 [185]. Artemisinin reduces apoptosis, suppresses cellular inflammation and inhibits OGD/R-induced oxidative stress and the NF- κ B signalling pathway in SH-SY5Y cells [190]. Triptolide inhibits inflammation and reduces apoptosis via the NF- κ B pathway [191]. Fraxin inhibits the PPAR γ /NF- κ B pathway to suppress oxidative stress, inflammatory response and apoptosis [192]. Didymine reduces inflammation and apoptosis by activating the PPAR pathway [193]. DI-3-n-butylphthalide (NBP), originally extracted from celery seeds, effectively reduced cognitive deficits, neuronal cell loss and apoptosis in mice. In addition, NBP promoted the expression of BDNF and TrkB, significantly inhibited TLR4/MyD88/NF- κ B signalling and suppressed microglial proliferation and production of pro-inflammatory mediators [194]. Schisandrin B suppresses TLR4-NF- κ B signaling circuit realization, reduces TLR4, NF- κ B and I- κ B phosphorylation, and cortical TNF- α , IL-6, IL-1 β and caspase-3 levels, while decreasing inflammation and apoptosis to promote neuronal repair [195].

Oroxyn A pretreatment activated the hedgehog signal pathway in a dose-dependent manner, increased Shh, p-smo and Gli2, and reduced apoptosis [182]. Leonurine prevents oxidative stress and neuronal apoptosis in cerebral ischemia

via the nitric oxide (NO)/nitric oxide synthase (NOS) system [196]. Baicalin (BC) combined with geniposide (GP) will downregulate microglial polarisation, inflammatory damage and neuronal apoptosis by inhibiting the 5-LOX/LTB4 pathway [197]. Nobiletin (NOB), an active natural flavonoid component of citrus, reduces neuronal apoptosis by inhibiting the Rho/ROCK pathway [198]. A decrease in RhoA/ROCK signaling-related protein expression, a decrease in Bax and caspase-3 expression, and an increase in Bcl-2 expression all contribute to Sanggenon C's beneficial effects on neurologic damage, brain edema, and cerebral infarction [187].

Through the NOD2/ MAPK/NF- κ B signaling pathway, asiaticoside decreases apoptosis by increasing Bcl-2 and decreasing Bax and caspase-3 expression [189]. By controlling the HIF-1 α /AKT/mTOR signalling pathway, naringin increases cell proliferation and decreases apoptosis and downregulates NF- κ B1, Bax and ROS while upregulating Bcl-2 [160]. Cardamonin protects against OGD/R-induced HBMEC apoptosis by activating the HIF-1 α /VEGFA pathway [199]. *Ligusticum striatum* DC. and borneol inhibit BMEC apoptosis, maintain BBB, and activate HIF-1 α /VEGF signaling pathway through synergistic action [200]. Naoluo Xintong Decoction targets angiogenesis and inhibits apoptosis by upregulating the HIF- 1 α /VEGF signaling pathway [201]. Astragaloside IV promotes autophagy and downregulates apoptosis by lowering cellular Bax/Bcl-2 levels and increasing LC3II/LC3I expression. This has neuroprotective benefits [202]. Plumbagin (5-hydroxy-2-methyl-1,4-naphthoquinone), a naphthoquinone derivative, also reduces MMP to inhibit neuronal apoptosis [203]. Green Tea Catechin reverses autophagy-activated microglia apoptosis via the AKT/mTOR signalling pathway [204]. Tao-hong Siwu Decoction inhibits both autophagy and apoptosis [205]. Silibinin balances the expression of Bcl-2 and Bax and inhibits apoptosis and autophagy [206]. Oridonin ameliorates caspase-9-mediated apoptosis in brain neurons by inhibiting RIPK3-mediated mitochondrial autophagy [207]. Echinacoside helps to improve cognitive impairment by inhibiting neuronal apoptosis and enhancing autophagy [208]. Morroniside, MOR is derived from *Cornus officinalis* and is a biologically active liposaccharide component [209]. Matrix metalloproteinases (MMPs) may perform protein hydrolytic activity and are linked to I/R damage, however this work showed for the first time that morroniside lowered MMP2 and MMP9, decreased caspase-3 expression, and enhanced the Bcl-2/Bax ratio [210].

Through modulation of the miR-375/SRXN1 axis, tea polysaccharide (TPS) was shown to be effective in preventing the death of astrocytes [211]. Celastrol inhibits apoptosis of human brain microvascular endothelial cells via circDLGAP4/miR-6085/GDF11 [212]. Vitexin Compound B-1 regulates miR-92b/NOX4 pathway and inhibits neuronal

apoptosis [213]. Ligustrazine targets the circ-0008146/miR-709/Cx3cr1 axis to inhibit apoptosis and inflammatory responses after cerebral ischemic injury [214]. Specifically, TPS suppressed miR-375 expression and greatly increased SRXN1 levels. Oleanolic acid (OA), suppressed OGD/R-induced PC12 cell inflammation and death via downregulating miR-142-5p [215]. MiR-23b controls the inflammatory substance pathway by targeting TAB3, an upstream positive regulator of the NF- κ B pathway. Chlorogenic acid increases miR-23b, and these results show that miR-23b plays a role in controlling apoptosis through the TAB3/NF- κ B/p53 axis; moreover, TAB3, NF- κ B, caspase-3, and p53 levels are all reduced [216]. This suggests that RNA may be used as a therapeutic target in IR. Isoastilbin (IAB) inhibits oxidative stress and neuronal apoptosis through the upregulation of SIRT1/3/6 expression [217]. Danggui-Shaoyao-San (DSS) inhibits apoptosis by activating the SIRT1 signalling pathway and suppressing NADPH oxidase [218]. Ovatodiolide (OVA), a bioactive substance extracted from *Anisomeles indica*, inhibits neuronal apoptosis and microglial inflammation via the SIRT1-NF- κ B pathway [219]. Baicalin inhibits hippocampal neuronal apoptosis by reducing the phosphorylation level of CaMKII [220]. Paeoniflorin inhibits neuronal apoptosis and promotes neural regeneration [221]. Osthole, a plant coumarin compound, inhibits apoptosis and protects neurons [222]. Sanguinarine inhibits inflammation and reduces apoptosis [223]. Salidroside inhibits RIP140-mediated inflammation and apoptosis [224]. The multifunctional gastrodin (GAS)-containing nano-drug carrier system has anti-inflammatory and anti-apoptotic effects on hypothalamic neurons, and the combination of herbal gastrodin with polymers prolongs the duration of drug circulation in the body and increases drug delivery to the target [225]. Betaine reduces cerebral infarct volume and neuronal apoptosis and increases the expression of antioxidant enzymes [226]. Dengzhan Xixin Injection protects against cerebral ischemia by reducing mitochondrial damage through the upregulation of mitochondrial phagocytosis and inhibiting mitochondria-mediated apoptosis [227]. Zuogui Pill (ZGP), a TCM, attenuates the effect of OGD/R on apoptosis and reduces nerve damage and infarct size [228]. Dihydromyricetin (DHM) is a flavonoid and regulates peri-infarct cortical apoptosis and astrocyte proliferation [229]. Icaritin (ICT) is a metabolite of icariin (ICA) that reduces I/R-induced neurological damage by decreasing neuronal apoptosis and senescence [230]. *Acanthopanax senticosus* (Rupr. & Maxim.) Harms (ASE) and *Gastrodia elata* Blume (GEB) have been used in traditional Chinese medicine. Their combination may improve neurological damage by reducing neuronal damage and preventing neuronal apoptosis by reducing oxidative stress and inflammation [231]. Moreover, the hedgehog pathway, the RhoA/ROCK pathway, the HIF, the NOD, the autophagy, the SIRT, HIF-1 α and other pathways can

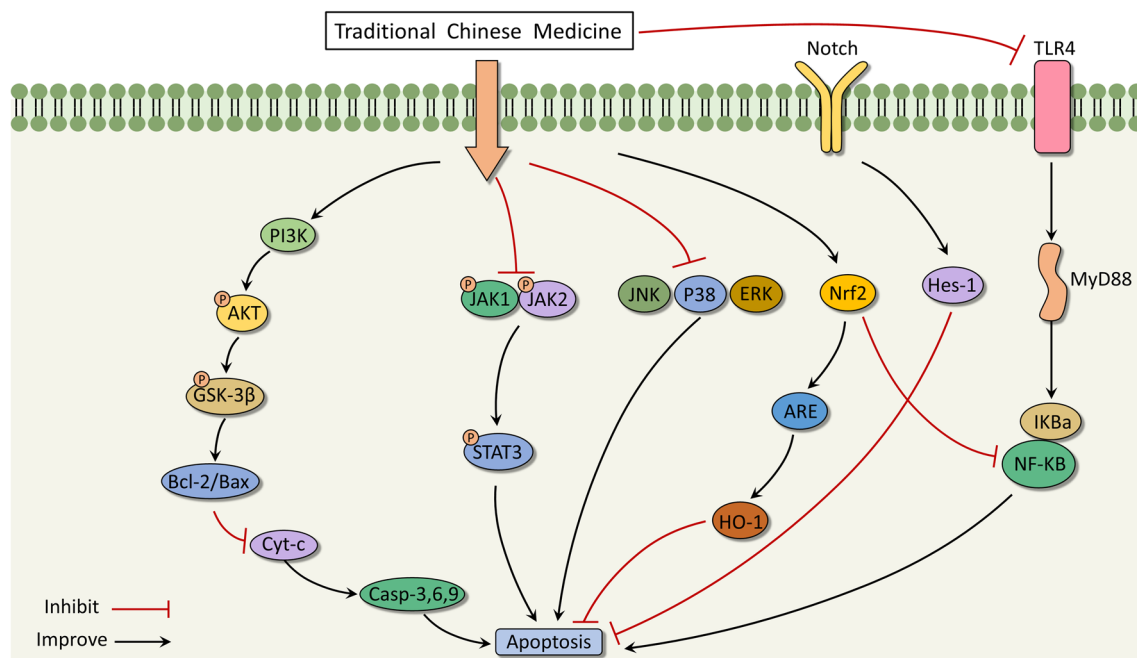


Fig. 5 Mechanism of Chinese medicine interfering with apoptosis of neuronal cells after CIR through cellular pathway

also reverse apoptosis, making the drug treatment pathway rich, where one of the factors fails, the therapeutic effect can be achieved through another pathway, as shown in the Fig. 5 for the specific mechanism. (Table 1)

Apoptosis-related proteins

Herbal therapies for CIR include proteins including p53, HSP70, AIF, Cyt-c, Smac/DIABLO, the Bcl-2 family including Bcl-2/Bax, PUMA, Bim and Bcl-xl, the inhibitor of apoptosis proteins (IAP) family including Survivin, Livin and X-linked IAP (XIAP) and others that are intimately connected to apoptosis and have a role in mediating these treatments.

P53 proteins pathway

To induce neuronal apoptosis, p53 activates caspase-3 and an intrinsic mechanism reliant on mitochondria [232]. Enhanced mitochondrial outer membrane permeability (MOMP), Cyt-c release, and ultimately death are all induced by wild-type p53 via interactions with Bcl-xl and Bcl-2 in response to stress [233]. Over-phosphorylated p53 eventually causes apoptosis, and after external insults such as brain ischemia or oxidative stress, p35 protein is converted to p25 protein, forming a CDK5-p25 protein complex, leading to over-activation of CDK5 and induction of phosphorylation of p53 protein [234, 235]. Salvianolic acid A inhibits the

activation of GSK3 β and p25/Cdk5 by activating PI3K/AKT signalling, which upregulates β -catenin and Bcl-2 and reduces neuronal apoptosis [236]. Apigenin has anti-apoptotic and antioxidant properties to reduce neuronal damage by down-regulating the expression of Nrf2 and P53 [237]. Tetramethylpyrazine (TMP) is a constituent of *Ligusticum chuanxiong Hort.* (chuan xiong), a traditional Chinese herb. TMP combined with borneol (BO) reduces P53 and inhibits autophagy and apoptosis in neuronal cells [238]. Because of its ability to suppress the NF- κ B/p53 signaling pathway, ginkgetin lowers Bax, p53, and Beclin 1 expression and greatly increases Bcl-2 expression in response to cerebral ischemia–reperfusion, hence preventing autophagy and apoptosis [181]. The combination of *Ligusticum chuanxiong Hort* Extract with Borneol suppressed calcium excess and reduced expression of p53, caspase-3, Bax, and Bcl-2 [239]. In mice with ischemic brain damage, cycloastragenol protected neurons by increasing SIRT1, blocking apoptosis and inflammation, lowering p53 acetylation and the Bax/Bcl-2 ratio, and blocking NF- κ B p65 nuclear translocation [240]. *Dracocephalum moldavica* reduced Bax protein but raised Bcl-2 and suppressed p53 gene expression [183]. Inhibiting apoptosis and promoting autophagy following SIRT1 activation, protocatechualdehyde upregulates SIRT1, Bcl-2, Beclin-1, and LC3 II/I expression while downregulating P53, P62, caspase3, and Bax [241]. The pro-apoptotic BH3-only protein not only binds to the Bcl-2 family of anti-apoptotic proteins and releases pro-apoptotic proteins like Bax, but it may also directly activate Bax and Bak, oligomerize them to

create mitochondrial membrane channels, and release cytochrome *c* from *Aglaia odorata* Lour. extract may inhibit the p53/puma-mediated mitochondrial caspase-9/3 apoptotic pathway has a significant neuroprotective effect on cerebral ischemia [186].

HSP70/AIF

There are multiple members of the heat shock protein 70 (HSP70) family, such as inducible HSP70 (HSP72), structural Hsc70 (HSP73), mitochondrial HSP75 (mt HSP75), and endoplasmic reticulum Grp78 (Bip). The HSP70 family alleviates ischemia via HSP70-dependent anti-apoptotic actions and lowers blood–brain barrier hyperpermeability and caspase-3 activity [242]. Hsp70 is a molecular chaperone protein that not only prevents apoptosis by binding to apoptotic protease-activating factor-1 (Apaf-1), hence preventing its activation of mitochondrial apoptosis, but also inhibits the AIF-mediated apoptotic pathway involving non-independent caspase way [243].

AIF is a caspase-independent apoptotic factor. AIF mediates two pathways of cell death, apoptosis induced by the hydrolysis of mature AIF to a 57 kDa truncated AIF by calpain, and the parthanatos pathway induced by the involvement of non-calpain processing of AIF [244, 245].

Enhanced AIF-Hsp70 interaction caused a decrease in AIF entry into the nucleus and attenuated apoptosis [246]. Danhong injection inhibited apoptosis by upregulating HSP70 and ameliorated oxidative stress, inhibited PARP1/AIF pathway activation, and thus reduced DNA damage [247]. Gualou Guizhi Decoction reduces the area of cerebral infarction while promoting the recovery of neuronal function, and its effects are related to the regulation of PARP-1/AIF pathway-related protein expression and translocation, inhibition of total protein expression of PARP-1, PAR, AIF and nucleic acid endonuclease G (Endo G) in the ischemic cortex, and a significant increase in total protein expression of Hsp70, thus exerting an anti-apoptotic effect [184]. Hsp70 may inhibit the translocation of AIF to the nucleus, hence preventing extensive DNA fragmentation and peripheral chromatin condensation in a caspase-independent way [246, 248]. The anti-apoptotic effect of Ferulic Acid is neuroprotective through down-regulation of Smac/DIABLO/XIAP/caspase-3 and activation of HSP70/Bcl-2 and HSP70/autophagy-induced signalling pathways [249].

Calpain-I resides in the mitochondrial membrane gap and is activated when an imbalance in Ca^{2+} homeostasis in the cell causes an increase in the intracellular Ca concentration, which in turn promotes the shearing of AIF and its release into the nucleus, where chromatin condensation and DNA damage occur, ultimately leading to apoptosis [250]. Baicalein can bind to the regulatory Calpain-I subunit binding, inhibits Calpain-I activation without regulating its expression and does not decrease calcium levels, but

inhibits nuclear translocation of AIF, thus improving IR by reversing disrupted ROS production, apoptosis and via the Calpain-I/AIF pathway [251].

Cytc protein

Serine-47 (S47) phosphorylation of Cytc in the brain results in lower cytochrome c oxidase (COX) activity and caspase-3 activity in vitro. Lack of Cytc phosphorylation leads to maximal electron transport chain flux, $\Delta\Psi\text{m}$ hyperpolarisation and ROS-triggered cell death. However, caspase-3 is a crucial protein for the induction of apoptosis, so intervening with Cytc could be a target for inhibiting apoptosis. It also confirms that the pro-survival kinase Akt (protein kinase B) is a likely mediator of the S47 phosphorylation of Cytc in the brain [252]. Hydroxysafflor yellow A reduced the opening of Cytc from mitochondria to inhibit mPTP export for its anti-I/R function, in addition to significantly improving neurological deficits and reducing infarct volume in rats [24]. Picoside II is a TCM that exerts neuroprotective effects by inhibiting the mitochondria-Cytc signalling pathway after ischemia–reperfusion injury in rats, scavenging ROS and reducing apoptotic cells [253]. Curdione attenuates Cytc, c-caspase-3 and c-caspase-9 and increases Bcl-2/Bax [254]. Mu-Xiang-You-Fang (MXYF) is a classic TCM used by Chinese minorities to treat disease. It also reduces Cytc, caspase-3 and caspase-9 [255]. Polydatin blocks Cytc translocation and impairs caspase-9 and caspase-3 activity [256]. Boropinol-B inhibits the hyperactivation of microglia and astrocytes and suppresses apoptosis by increasing the expression of Bcl-2 and decreasing the expression of Bax, caspase-3 and Cytc [257].

Smac/DIABLO protein

Smac/DIABLO can bind to IAPs and weaken the inhibitory effect of IAPs on caspases, thereby inhibiting cell proliferation and promoting apoptosis. The leaves of the Khat shrub include the major alkaloid compounds (cathinone) and cathine. Khat treatment can induce apoptosis and exacerbate neurological damage by enhancing the release of Smac/DIABLO from the mitochondria into the cytoplasm and should therefore be avoided in people with cerebrovascular disease [258]. Xiao-Xu-Ming decoction [21], Ferulic Acid [249], and *Alpinia oxyphylla* Miq [109] also down-regulate Smac/DIABLO and thus exert anti-apoptotic effects.

Bcl-2/Bax protein

Protodioscin down-regulates Bax and up-regulates Bcl-2, thus inhibiting neuronal apoptosis and reducing the release of inflammatory factors [259]. Daphnetin (DAP) is a coumarin derivative extracted from *Daphne odora* var. DAP

inhibits TLR4/NF- κ B expression and reduces neuronal apoptosis [260]. Protosapannin A selectively induces the degradation of the pro-apoptotic protein Bax without affecting other Bcl-2 family members, and in addition maintains mitochondrial function in neuronal cells through Bax/LC3B binding, which in turn promotes the autophagic degradation of Bax [261]. Astragaloside IV reduces brain I/R injury in rats by increasing Cx36 and PKA protein expression and decreasing the Bax/Bcl-2 ratio [262]. Vulgarisin-type diterpenoids from self-heal (*Prunella vulgaris*) significantly inhibit apoptosis by regulating Bax/Bcl-2 protein expression and inhibiting the expression of cleaved caspase-3 and caspase-9 [263]. Skullcapflavone II (SCP), a flavonoid isolated from *S. baicalensis*, increases Bcl2 and induces angiogenesis while reducing Bax and NF- κ B/TLR4, inhibiting apoptosis [264]. N-butyl alcohol extracts from *Rhizoma Pinelliae Pedatisectae* against cerebral ischemia–reperfusion injury play neuronal anti-apoptosis [265]. Several drugs also inhibit apoptosis through regulation of the Bcl-2/Bax ratio, including β -Caryophyllene [266], Qiancao Naomaitong Mixture (QNM) [267], Sesamol [268], 20(R)-Ginsenoside Rg3 [269], Oleuropein, a natural extract from plants [270], Gastrodin (GAS) extracted from the Chinese herbal medicine *Gastrodia elata* Blume [271].

PUMA protein

PUMA, a BH3-only member of the Bcl-2 protein family, is also required for both p53-dependent and p53-independent forms of apoptosis. In addition, knockdown of PUMA under oxygen, glucose and serum-deprived culture conditions resulted in inhibition of Bax, cleaved caspase-9 and caspase-3 expression [272]. MiR-211 overexpression attenuates OGD/R-induced apoptosis in PC12 cells by regulating PUMA [273]. Triptolide is a compound extracted from the Chinese herb thunder god vine (*Tripterygium wilfordii* Hook F.). It down-regulated the neuroprotective effects of NF- κ B, PUMA and caspase-3 on CIR injury in rats by inhibiting NF- κ B/PUMA signalling [274]. *Aglaia odorata* Lour. Extracts may inhibit the p53/puma-mediated mitochondrial caspase-9/3 apoptotic pathway and have a significant neuroprotective effect on cerebral ischemia [186].

Bim protein

Bim deficiency modulates post-stroke outcomes, including brain inflammation and reduced leukocyte apoptosis after systemic stroke; therefore, Bim could be a potential therapeutic target for stroke [275]. Salvianolic acid A inhibits neuronal apoptosis through the inhibition of the AKT/FOXO3a/BIM pathway [276].

EnDoG protein

Since EnDoG acts as a nucleic acid endonuclease, it can cleave both DNA and RNA, and it is translocated from the mitochondria to the nucleus, where it may take part in chromatin degradation with other nuclear proteins [277]. Picroside II prevents mitochondrial permeability transition pore (mPTP) permeability and EnDoG release while downregulating VDAC1, as discovered by Li et al. [278].

XIAP protein

XIAP directly regulates and inhibits the activity of caspase-3, caspase-7 and caspase-9 and exerts anti-apoptotic effects [279]. The Xiao-Xu-Ming decoction raises c-IAP1, lowers p53 and Bax levels, and enhances Bcl-2 [21]. The anti-apoptotic effect of Ferulic Acid is neuroprotective through down-regulation of Smac/DIABLO/XIAP/caspase-3 [249].

Survivin protein

The anti-apoptotic protein Livin has been investigated less in cerebral ischemia and exclusively in tumor cells like glioblastoma and astrocytoma, while Survivin can control both apoptosis and cell differentiation. IMM-H004 is a coumarin-derived 3-piperazinylcoumarin compound that promotes the expression of the anti-apoptotic protein survivin, induces FoxO1 phosphorylation, increases the expression of hepatitis B x-interacting protein (HBXIP), and HBXIP forms a complex with survivin to prevent caspase death activation of the cascade, thereby preventing the death of apoptotic cells [280].

Discussion and conclusion

Although apoptosis is the earliest known pathway of cell death, its combination with Chinese medicine for the development of small molecule clinical therapeutics is a relatively late start. Therefore, based on apoptosis occurring after cerebral ischemia–reperfusion injury, this paper reviews recent findings and advances in Chinese medicine intervention on apoptosis. The mechanism of action of one or two active ingredients of Chinese medicine, Chinese medicine injections and Chinese medicine compound against CIR is explored from three aspects: classical endogenous and exogenous apoptotic pathways, apoptosis-related signalling pathways and apoptosis-related proteins. Endogenous apoptotic pathways mediated by organelles mitochondria and endoplasmic reticulum, exogenous apoptotic pathways mediated by death receptors, apoptosis-related signalling pathways such as PI3K/AKT, MAPK/ERK, JAK/STAT, AMPK, and

apoptosis-related proteins such as Bcl-2 family, XIAP family, P53, AIF are reviewed in detail. This demonstrates the multiple mechanisms of anti-apoptosis in Chinese medicine. Moreover, Chinese medicine usually interferes with apoptosis while ameliorating oxidative stress, reducing reactive oxygen species and inflammation levels, and interacting with autophagy to exert anti-CIR cell damage. A review found that the number of studies on anti-apoptosis in Chinese medicine is relatively large, and most of them are at the stage of cellular and animal models.

The clinical and experimental applications of Chinese herbal medicines have characteristics such as the natural compound form of the active ingredients of Chinese herbal medicines has low bioavailability and intestinal absorption, and the presence of multiple components in Chinese herbal tonics helps to enhance the efficacy of CIR. However it also makes it difficult to control the quality or safety of herbal medicines. The number of neuroprotective agents or drugs available for clinical use is minimal [281], and the presence of potent small molecules in TCM is not readily accessible across the blood–brain barrier to the site of cerebral ischemia, making the combination of nanotechnology or exosomes with TCM to enhance the therapeutic effect.

In addition, there is a lack of multi-pathway crossover studies, such as the synergistic effects of TCM on different signalling pathways, whether and how apoptosis and other programmed cell deaths, such as iron death or cell scorching, are regulated simultaneously, and whether there is crosstalk between the two and the specific mechanism, which warrants a more comprehensive understanding of the mechanism of TCM in regulating apoptosis and anti-CIR in vivo.

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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